HESOD: HIGH-RESOLUTION EFFICIENT SMALL OBJECT DETECTION VIA SEMANTIC-SPECTRAL DUAL SELECTION

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ABSTRACT

Detecting tiny objects in high-resolution aerial and maritime imagery is severely challenged by background redundancy and extreme scale variation. While spatial selective computation routes foreground regions to cut latency, existing single-evidence selectors suffer from severe semantic collapse on micro targets, causing irrecoverable coarse-stage pruning. We propose **HESOD** (**H**igh-Resolution **E**fficient **S**mall **O**bject **D**etection), a dual-evidence framework featuring semantic-spectral routing, coverage-constrained spatial optimization, and lightweight downstream detection. HESOD couples semantic objectness with constant-width spectral saliency under a monotonic evidence-preserving max-fusion rule to rescue degraded targets. To promote recall-safe routing, an object-level soft-coverage loss directly optimizes multi-cell joint instance coverage across ground-truth objects. To recover downstream efficiency, an inverted residual decoupled head (ISPP) is integrated to compress parameters by $> 27\%$ with only fractional FLOPs overhead. Extensive experiments show that HESOD improves $\mathrm{mAP@.5}$ by 0.9 pp and 2.3 pp (boosting total recall by 3.66 pp and 3.33 pp) on UAVDT and SeaPerson, respectively, establishing an effective accuracy-efficiency co-design suitable for resource-constrained deployment.

Index Terms— Small object detection, selective computation, spectral saliency, coverage optimization, UAV

1. INTRODUCTION

High-resolution aerial and maritime vision is fundamental to intelligent surveillance, environmental monitoring, and search-and-rescue. In operational scenarios across aerial and maritime domains [1–3], targets of interest typically occupy only a handful of pixels ($< 16 \times 16$ px) amid massive uniform backgrounds ($> 70\%$–80%) [4, 5]. Evaluating standard dense deep detectors across multi-megapixel grids (1280×1280 to 2048×2048) incurs prohibitive arithmetic latency on resource-constrained platforms.

To eliminate redundant computation, *spatial selective computation* dynamically routes feature processing strictly to prospective foreground regions [4, 6, 7]. Notably, ESOD [4] predicts an early spatial priority heatmap directly after a shallow stem, slicing features into candidate bounding patches (AdaSlicer) and bypassing background across downstream stages.

Crucially, spatial selection operates under a severe *asymmetric error cost*: false-positive background patches merely incur minor compute overhead, whereas discarding a true tiny target produces an *irrecoverable false negative* across downstream stages [4]. Existing detectors like ESOD [4] rely exclusively on single semantic objectness with independent per-cell BCE. Under extreme downsampling, micro targets suffer severe semantic collapse, causing faint instances to be prematurely pruned (Fig. 1(b)). To overcome this recall dilemma, we observe that tiny targets retain prominent high-frequency spatial gradients and spectral saliency even when deep semantics fade [8]. By coupling dual-evidence routing with an object-level soft-coverage loss, our selector reliably rescues dropped targets (Fig. 1(c)). We propose **HESOD** (**H**igh-Resolution **E**fficient **S**mall **O**bject **D**etection), a dual-evidence framework uniting semantic-spectral routing, coverage-constrained spatial optimization, and lightweight downstream detection. More specifically, our key contributions are:

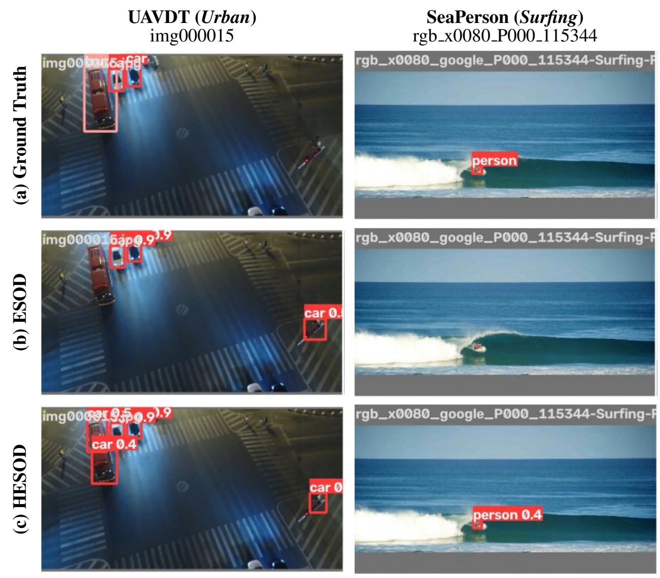

Fig. 1. Visual comparison on UAVDT and SeaPerson. Baseline ESOD drops micro targets (b), while HESOD restores full instance recall (c).

1. **Semantic-Spectral Dual Selection**: We couple semantic objectness with constant-width spectral filtering, independent of backbone channel width C, extracting directional gradients to rescue micro targets from semantic collapse.
2. **Evidence Preservation and Soft Coverage**: We formulate a monotonic max-fusion rule and an object-level soft-coverage loss to directly optimize multi-cell instance recall under irreversible pruning.
3. **Lightweight Downstream Co-Design**: We integrate an ISPP-style lightweight decoupled head [15] downstream of the selective router, slashing detector parameters by $> 27\%$ and recovering routing compute overhead with only fractional FLOPs overhead.

Source code and weights will be available upon acceptance.

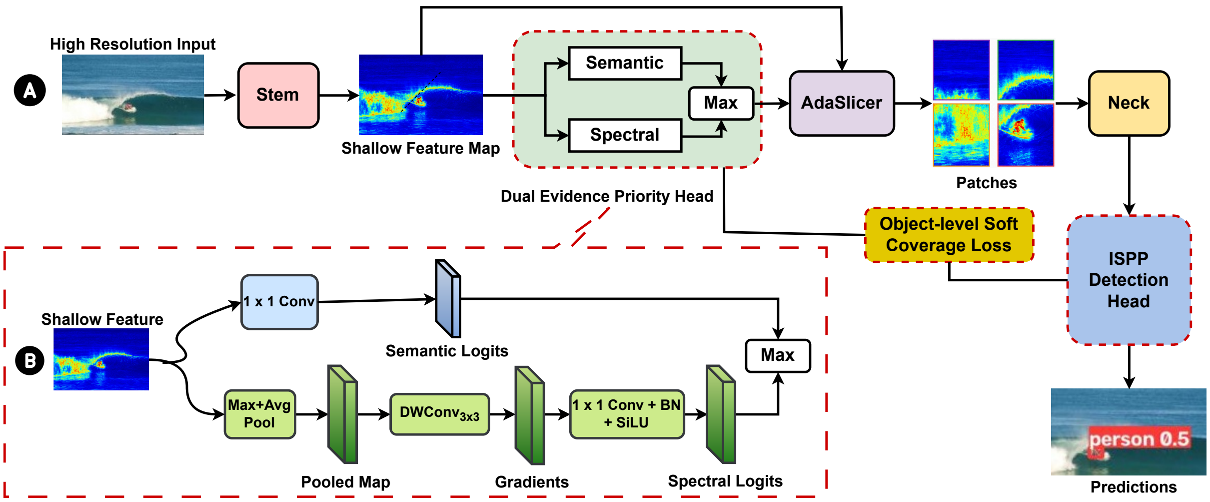

Fig. 2. Overview of HESOD: (a) End-to-end selective pipeline; (b) Dual-Evidence Priority Head. Dashed boxes highlight our proposed contributions over ESOD.

2. RELATED WORK

2.1. Small Object Detection in High-Resolution Vision

Detecting tiny targets in high-resolution aerial and maritime imagery is severely challenged by extreme scale disparity, low contrast, and massive background areas ($> 70\%$–80%) [1–3, 9]. Classical approaches utilize density-guided cropping (ClusDet [10], DMNet [11]) or scale-adaptive label assignment (NWD [12], RFLA [13]). Recently, ResBiDet [14] introduced a coarse-to-fine framework using global heatmap guidance for region cropping. However, it relies on multi-scale image-level slicing and boundary stitching. In contrast, HESOD performs early feature-level selective routing directly within the detector trunk and explicitly optimizes instance coverage under irreversible pruning.

2.2. Selective and Dynamic Computation

Dynamic computation accelerates inference by allocating compute strictly to prospective foreground regions (e.g., AutoFocus [5], QueryDet [6], CEASC [7]). Notably, ESOD [4] predicts an early-stage spatial priority heatmap to extract bounding patches (AdaSlicer). However, ESOD relies exclusively on a single semantic score supervised by independent per-cell BCE. Under extreme scale disparity, micro targets suffer from semantic feature collapse after downsampling; meanwhile, per-cell BCE lacks joint instance-level coverage constraints, causing fatal false negatives under an asymmetric error cost. HESOD resolves this by combining semantic and spectral dual cues with object-level soft-coverage optimization.

2.3. Spectral Saliency and Frequency Methods

Frequency-domain representations offer rich structural priors for micro targets, which retain sharp high-frequency edge gradients even when photometric semantics fade [8]. While SET [8] applies dense frequency transforms across multi-scale pyramids with prohibitive latency, HESOD repurposes spectral saliency as an ultra-lightweight routing cue: channel pooling compresses stem features to 2 channels, enabling directional Laplacian and Sobel gradient filtering at constant width, independent of backbone channel dimension C.

3. PROPOSED METHOD

The architecture of HESOD is shown in Fig. 2(a). From input $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, a shallow stem extracts dense features $\mathbf{F} \in \mathbb{R}^{H_s\times W_s\times C}$. The *Dual-Evidence Spatial Selector* then coordinates: (i) a priority head fusing semantic objectness with channel-pooled spectral saliency (Fig. 2(b)) into spatial priority map $\mathbf{S} \in [0,1]^{H_s\times W_s}$; and (ii) an AdaSlicer module dynamically clustering candidate foreground cells into bounding patches $\mathcal{P}$ from $\mathbf{F}$. Downstream backbone stages and lightweight decoupled heads evaluate only retained patches $\mathcal{P}$ under soft-coverage supervision (§3.2, §3.3).

3.1. Dual-Evidence Priority Estimation

To rescue micro targets with degraded semantic activations, HESOD couples high-level semantic confidence with low-level structural gradient saliency at negligible compute overhead (Fig. 2(b)).

Let $\mathbf{f}_i \in \mathbb{R}^C$ denote the feature vector at spatial location i in $\mathbf{F}$. The semantic branch predicts class-specific logits via a 1×1 convolution: $\mathbf{z}_i^{\text{sem}} = \text{Conv}_{1\times 1}(\mathbf{f}_i) \in \mathbb{R}^{N_c}$. In parallel, the spectral branch extracts high-frequency structural gradients by first compressing channels via max- and mean-pooling:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:,:,c} \,\|\, \frac{1}{C}\sum_{c=1}^{C}\mathbf{F}_{:,:,c}\right] \in \mathbb{R}^{H_s\times W_s\times 2}, \quad (1)$$

where $[\cdot \| \cdot]$ denotes channel concatenation. Compressing $\mathbf{F}$ to 2 channels retains peak edge activation and background energy while ensuring $\mathcal{O}(H_sW_s)$ spatial filtering complexity independent of backbone width C. Next, a learnable depthwise 3×3 convolution bank—initialized with discrete Laplacian (isotropic second-order edge transitions) and directional horizontal/vertical Sobel operators—filters $\mathbf{F}_{\text{pooled}}$ into 6 structural channels. A 1×1 projection layer maps the filtered channels to dimension N_c, yielding spectral logits $\mathbf{z}_i^{\text{spec}} \in \mathbb{R}^{N_c}$.

To prevent confidence dilution on weak targets, evidence fusion enforces *evidence preservation* via a parameter-free elementwise maximum:

$$z_{i,c} = \max\left(z_{i,c}^{\text{sem}},\, z_{i,c}^{\text{spec}}\right), \quad s_i = \max_{c=1}^{N_c} \sigma(z_{i,c}), \quad (2)$$

where taking the elementwise maximum guarantees monotonic evidence preservation ($z_{i,c} \geq z_{i,c}^{\text{sem}}$ and $z_{i,c} \geq z_{i,c}^{\text{spec}}$) at every spatial

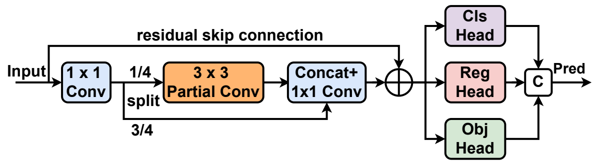

Fig. 3. Inverted Residual Decoupled Head (ISPP) with Partial Convolution (Ⓒ denotes channel concatenation).

grid cell. Candidate cells with routing priority $s_i > \tau$ are clustered into bounding patches $\mathcal{P}$ and forwarded downstream.

3.2. Object-Level Soft-Coverage Loss

Standard pixel-wise binary cross-entropy (BCE) evaluates grid cells independently, failing to guarantee instance-level coverage: a selector may over-allocate cells to large objects while dropping micro objects whose receptive fields contain only weak responses. To prevent premature false negatives, we formulate a differentiable object-level soft-coverage loss.

Let $\mathcal{N}(j)$ denote the set of selector cells overlapping ground-truth box $j \in \{1, \dots, N_{\text{gt}}\}$. We formulate a noisy-OR-inspired differentiable coverage surrogate for instance j being covered by at least one selected cell:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i). \tag{3}$$

The per-instance coverage loss is defined as:

$$\ell_j = -w_j \log \left(p_j^{\text{cover}} + \epsilon \right), \tag{4}$$

where $w_j = \text{clip}(4/a_j, 1, 5)$ is a scale-adaptive weight based on bounding box area a_j in cells, penalizing misses on micro targets ($a_j < 4$). For any cell $i \in \mathcal{N}(j)$, the gradient of ℓ_j with respect to cell score s_i is:

$$\frac{\partial \ell_j}{\partial s_i} = -\frac{w_j}{p_j^{\text{cover}} + \epsilon} \prod_{k \in \mathcal{N}(j) \setminus \{i\}} (1 - s_k). \tag{5}$$

When neighboring candidate cells have low scores ($s_k \to 0$), the product term approaches 1, generating a strong gradient to boost cell i and encouraging at least one cell to activate per instance. The overall soft-coverage loss averages over instances: $\mathcal{L}_{\text{cover}} = \frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} \ell_j$. The multi-task training objective is $\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}(\mathcal{L}_{\text{BCE}} + \lambda_{\text{cov}} \mathcal{L}_{\text{cover}})$, with $\lambda_{\text{sel}} = 0.2$ and $\lambda_{\text{cov}} = 0.5$.

3.3. Lightweight Decoupled Head and Training Protocol

While dual-evidence routing substantially improves patch coverage by retaining faint foreground regions, evaluating more candidate patches increases downstream detector computation. To recover this overhead without sacrificing selector gains, HESOD incorporates an inverted residual partial-convolution (ISPP) decoupled head [15] as a dedicated compute compensator.

As shown in Fig. 3, the ISPP head expands neck feature channels to $2C_1$, partitioning them into an active branch (0.25 ratio) processed by a 3×3 Partial Convolution (PConv) and an identity passthrough branch (0.75 ratio). After linear projection and residual addition, decoupled 1×1 heads predict classification, box regression, and objectness independently, lowering arithmetic complexity while eliminating inter-task gradient interference.

Two-Stage Training Protocol. To ensure stable routing behavior, HESOD employs a two-stage training scheme: (i) *Stage 1 (Selector Convergence)*: Train the Dual-Max selector with a coupled head under $\mathcal{L}_{\text{cover}}$, establishing a well-calibrated priority heatmap; and

Table 1. Benchmark results on UAVDT and SeaPerson.

Method	mAP @.5	Params (M)	GFLOPs	Total Recall	BPR	FPS
UAVDT Benchmark *(1280 × 1280)*						
Faster R-CNN [16]	0.336	43.3	772.5	79.68%	–	18.8
RetinaNet [17]	0.296	36.4	360.9	83.44%	–	19.8
ESOD [4]	0.385	35.9	**68.2**	85.17%	0.884	**117.8**
HESOD (Ours)	**0.394**	**26.0**	74.9	**88.83%**	**0.940**	106.1
Δ *vs. ESOD*	**+0.9 pp**	**-27.5%**	+9.8%	**+3.66 pp**	**+5.6 pp**	-9.9%
SeaPerson Benchmark *(2048 × 2048)*						
Faster R-CNN [16]	0.551	43.3	1546.8	67.27%	–	23.1
RetinaNet [17]	0.473	36.4	942.7	65.98%	–	28.0
ESOD [4]	0.750	35.8	**202.4**	84.42%	0.947	**85.7**
HESOD (Ours)	**0.773**	**25.9**	208.5	**87.75%**	**0.991**	80.6
Δ *vs. ESOD*	**+2.3 pp**	**-27.6%**	+3.0%	**+3.33 pp**	**+4.4 pp**	-6.0%

BPR: Bounding Patch Recall. **Bold**: best; underline: second-best (selective models).

(ii) *Stage 2 (Head Fine-Tuning)*: Freeze the upstream trunk (backbone, dual branches, router) and fine-tune the lightweight ISPP neck and decoupled heads on the frozen routing distribution. Relative to the uncompressed Dual-Max selector, this lightweight head slashes GFLOPs by 16.9% and parameters by 27.5% (on UAVDT) while fully preserving peak coverage (0.940 BPR), maintaining competitive real-time FPS with only fractional overhead vs. ESOD.

4. EXPERIMENTS

4.1. Experiment Settings

We evaluate HESOD on two demanding high-resolution small-object benchmarks: **UAVDT** [2] (1280×1280 px, urban vehicle surveillance across 3 classes) and **SeaPerson** [3,9] (2048×2048 px, maritime search-and-rescue with $> 85\%$ micro targets < 32 px). Both evaluate full-resolution images without artificial cropping.

All models employ YOLOv5m in PyTorch and train for 50 epochs under selective protocols [4]; HESOD trains the selector in Stage 1 and fine-tunes the lightweight ISPP head with the frozen selector in Stage 2 to maintain routing stability. Metrics include COCO mAP@.5, scale-stratified recall across bins (*Very Tiny* $< 16^2$, *Tiny* 16^2–32^2, *Small* 32^2–96^2), and *Bounding Patch Recall* (BPR $= N_{\text{cov}}/N_{\text{gt}}$, where a target is covered if $|\text{GT} \cap \mathcal{P}|/|\text{GT}| > 0.5$ by selected patches $\mathcal{P}$). Latency (FPS) is measured on an NVIDIA RTX 5090 GPU at batch size 1 with FP32 precision.

4.2. Main Benchmark Comparisons

Comparison with Dense Baselines. As shown in Table 1, dense detectors incur prohibitive computational overhead on multi-megapixel grids. On **UAVDT**, HESOD outperforms Faster R-CNN and RetinaNet by **+5.8 pp** and **+9.8 pp** mAP@.5, slashing GFLOPs by **10.3**×/**4.8**× at **5.5**× higher FPS (106.1 vs. 18.8/19.8) and raising Very Tiny recall to **80.76%** (vs. 70.35%/72.77%). On **SeaPerson**, HESOD boosts mAP@.5 by **+22.2 pp/+30.0 pp** while cutting GFLOPs by **7.4**×/**4.5**× at **80.6** FPS, lifting recall from 67.27%/65.98% to **87.75%** (Very Tiny: 46.07%/50.05% → **75.71%**), confirming the necessity of selective computation.

Comparison with Selective Baseline (ESOD). Against ESOD, HESOD achieves substantial recall recovery and parameter compression with only fractional FLOPs overhead (Table 1). On **SeaPerson** ($> 85\%$ targets < 32 px), HESOD boosts mAP@.5 by **+2.3 pp** (0.750 → **0.773**) and BPR by **+4.4 pp** to **0.991**

Table 2. Module ablation, efficiency, and scale recall breakdown on UAVDT and SeaPerson.

	Detection Accuracy and Computational Efficiency						**Physical-Scale Recall Breakdown**				
Configuration	**mAP @.5**	**mAP @.5:.95**	**BPR**	**Params (M)**	**GFLOPs**	**FPS**	**Very Tiny** ($< 16^2$)	**Tiny** (16^2–32^2)	**Small** (32^2–96^2)	**Med/Lrg** ($> 96^2$)	**Total Recall**
UAVDT Benchmark *(*1280×1280*)*							(74,375)	(206,015)	(86,282)	(7,325)	(373,997)
(1) ESOD Baseline*	0.385	0.214	0.884	35.9	**68.2**	**117.8**	79.43%	84.21%	94.54%	59.78%	85.17%
(2) Semantic-only	0.384	0.217	0.906	35.9	75.0	113.8	79.13%	83.91%	94.00%	59.95%	84.82%
(3) Spectral-only	0.387	0.209	0.942	36.0	93.7	102.3	82.43%	88.72%	96.66%	66.80%	88.87%
(4) (3) + Channel-Pooled	0.394	0.209	**0.963**	35.9	99.3	97.6	**86.88%**	**92.23%**	**97.74%**	66.43%	**91.93%**
(5) Dual-Concat	0.371	0.205	0.919	35.9	83.9	107.2	79.09%	85.85%	95.85%	62.66%	86.35%
(6) Dual-Max	**0.395**	**0.218**	0.940	35.9	90.1	102.3	85.69%	89.97%	97.37%	65.99%	90.36%
(7) **HESOD (Ours)**	0.394	0.215	0.940	**26.0**	74.9	106.1	80.76%	88.98%	97.11%	**69.16%**	88.83%
SeaPerson Benchmark *(*2048×2048*)*							(82,417)	(174,816)	(42,987)	(155)	(300,375)
(1) ESOD Baseline*	0.750	0.320	0.947	35.8	**202.4**	**85.7**	74.03%	86.78%	94.74%	79.35%	84.42%
(2) Semantic-only	0.769	0.325	0.991	35.8	266.8	73.5	75.49%	91.57%	95.71%	81.94%	87.75%
(3) Spectral-only	0.770	0.327	**0.992**	35.8	267.4	71.7	75.23%	91.69%	**95.89%**	86.45%	87.77%
(4) (3) + Channel-Pooled	0.767	0.324	0.988	35.8	263.4	72.7	**76.13%**	91.36%	95.50%	**87.10%**	87.77%
(5) Dual-Concat	0.772	0.326	0.986	35.8	281.2	69.8	76.00%	91.50%	95.68%	80.00%	87.84%
(6) Dual-Max	**0.778**	**0.330**	0.991	35.8	255.1	73.8	75.65%	**92.06%**	95.85%	81.94%	**88.10%**
(7) **HESOD (Ours)**	0.773	0.327	0.991	**25.9**	208.5	80.6	75.71%	91.47%	95.72%	80.65%	87.75%

*Trained with standard BCE (others use $\mathcal{L}_{\text{cover}}$). Parenthetical values denote ground-truth counts per bin. One-to-one matching at conf ≥ 0.001, IoU ≥ 0.5. **Bold**: best; underline: second-best.

(total recall $+3.33$ pp to **87.75%**), while compressing parameters by **27.6%** (25.92M vs. 35.78M) at 80.6 FPS. On **UAVDT** (1280×1280), HESOD improves mAP@.5 to **0.394** (+0.9 pp) and BPR to **0.940** (+5.6 pp, rescuing +13,700 targets), cutting parameters by **27.5%** at 106.1 FPS (+3.66 pp total recall) with class-wide recall gains: *car* (85.6% → 89.0%), *truck* (81.1% → 86.5%), and *bus* (67.7% → 81.4%, **+13.7 pp**).

4.3. Ablation Studies and Scale Analysis

Evidence Preservation Principle. As reported in Table 2, unconstrained affine concatenation (Arm 5, 0.371 mAP) degrades performance because unconstrained linear combinations dilute confident cues. In contrast, parameter-free Elementwise Max (Arm 6) enforces monotonic evidence preservation ($z_{i,c} \geq \max(z_{i,c}^{\text{sem}}, z_{i,c}^{\text{spec}})$), achieving peak selector accuracy of **0.395** mAP on UAVDT and **0.778** mAP on SeaPerson (0.991 BPR) while maintaining stable high-IoU localization (mAP@.5 : .95 of 0.218 and 0.330).

Efficiency Recovery via Lightweight Head. The inverted residual ISPP head serves as a dedicated compute compensator. On UAVDT, integrating ISPP (Arm 7) preserves 0.394 mAP@.5 and 0.940 BPR while slashing GFLOPs by 16.9% (90.1 → 74.9) and parameters by **27.5%** at 106.1 FPS relative to Dual-Max. On SeaPerson, ISPP similarly cuts GFLOPs by **18.2%** (255.1 → 208.5) and parameters by **27.6%** (35.79M → 25.92M) with BPR fully preserved at 0.991, raising FPS from 73.8 to 80.6 (+9.2%) with minimal accuracy trade-off (0.778 → 0.773 mAP).

Micro-Scale Recall Recovery. In the scale breakdown of Table 2, baseline ESOD suffers severe degradation on *Very Tiny* targets ($< 16^2$ px), dropping to 79.43% on UAVDT and 74.03% on SeaPerson due to early semantic collapse. Dual-Max routing restores extreme-scale sensitivity across both domains, boosting *Very Tiny* recall by **+6.26 pp** on UAVDT and improving *Tiny* recall by **+5.28 pp** on SeaPerson (92.06% vs. 86.78%, with Very Tiny recall reaching **75.71%**).

Channel Pooling Benefit. Crucially, channel pooling demonstrates its decisive advantage when integrated into the unified Dual-Max selector. On SeaPerson, full-width spectral filtering suffers severe memory pressure (forcing training batch size to 2 due to VRAM OOM) and reaches only 0.763 mAP@.5 at 277.8 GFLOPs. Compressing features to 2 channels enables standard batch-8 training, lifting mAP@.5 by **+1.5 pp** (0.763 → **0.778**), boosting BPR to **0.991**, and cutting GFLOPs by **8.9%** (277.8 → 255.1) with 5.0% higher FPS. Single-branch diagnostics (Arms 3 vs. 4) further confirm that channel pooling suppresses intra-channel noise and lifts *Very Tiny* recall across both benchmarks (+4.45 pp on UAVDT, +0.90 pp on SeaPerson).

Error Bottleneck Diagnosis. Error analysis reveals that dual-evidence routing and coverage optimization slash selector false negatives from 25.6% to 19.2% on SeaPerson (achieving a near-flawless **99.1%** BPR) and rescue over 13,700 dropped instances on UAVDT (BPR 0.884 → **0.940**), virtually eliminating early selector dropouts and shifting the remaining challenge to downstream regression refinement.

Latency and Qualitative Profiling. On an RTX 5090 GPU, stem extraction takes 1.2 ms, Dual-Evidence selection 1.8 ms, AdaSlicer 0.9 ms, and ISPP 5.5 ms (106.1 FPS on UAVDT, 80.6 FPS on SeaPerson). As visualized in Fig. 1, HESOD accurately localizes faint micro targets missed by ESOD without generating false alarms.

5. CONCLUSION

In this paper, we presented HESOD, an efficient dual-evidence framework for small object detection in high-resolution vision. By uniting semantic-spectral routing, evidence preservation, and object-level soft coverage ($\mathcal{L}_{\text{cover}}$), HESOD prevents premature selector pruning of micro targets. Coupled with a lightweight decoupled head (ISPP), HESOD improves mAP@.5 by 0.9 pp and 2.3 pp (total recall +3.66 pp and +3.33 pp) on UAVDT and SeaPerson, respectively, with $> 27\%$ fewer parameters and only fractional FLOPs overhead, achieving an effective accuracy-efficiency co-design for resource-constrained deployment. Future work will extend dual-evidence selective routing to multimodal perception platforms.

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